

Wu Xiaoyun

Graduating in August 2026

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Professional Experience

Backend Software Engineer Intern - Global LIVE Wallet [TikTok](#) **Singapore** 12/2024 - 08/2025

- Optimized core RPC services by rewriting high-QPS handlers (4.4 million peak QPS) from [Golang](#) to [Rust](#), achieving 50% compute resource savings and 30% latency reduction; collaborated extensively with the Rust service framework team, conducted thorough staging/production testing and traffic diversion, projecting \$300K in annual server cost savings and freeing resources for critical adjacent projects.
- Enhanced on-call efficiency for Wallet services by developing AI-driven analysers and auto-conclusion features in the Oncall Agent; utilised Golang, LLMs, and vector databases to rank relevant events, correlate error logs with code locations, and suggest root causes, significantly reducing engineers' on-call burden and improving overall development efficiency.

Frontend Software Engineer Intern - Global LIVE Money Platform [TikTok](#) **Singapore** 05/2024 - 08/2024

- Reduced CI/CD pipeline and production deployment times by 33% (from 15 to 10 minutes) by migrating the Money Platform from [Webpack](#) to [Rspack](#), doubling asset bundling speed. Drove cross-functional collaboration with Rspack, [Rsbuild](#), and TikTok LIVE Solutions teams.
- Improved Money Platform quality by creating a white-screen test summary page and upgrading the white-screen alarm Lark bot, enhancing debugging with detailed reports including console logs and network requests viewable in Chrome DevTools—eliminating local bug reproduction and accelerating alarm resolution.
- Developed an alarm Lark bot to streamline workflows by aggregating, filtering, and analysing alarms; designed a standardised API for seamless third-party integration and incorporated white-screen alarms into this framework.

Software Engineer Intern [Computing for Voluntary Welfare Organisations](#) **Singapore** 05/2023 - 08/2023

- Improved backend throughput by 5 times by migrating a Case Management System backend from Ruby on Rails to Golang. Notable features ported include RBAC, Auth, 2FA and Masquerading.

Projects

[RustScript](#)

- Engineered RustScript, a statically-typed language featuring a custom compiler and VM; implemented type-checking, concurrency primitives, and bytecode execution, achieving performance parity with similar interpreted languages.

Education

BSc Computer Science, second major in Statistics [National University of Singapore](#) **Singapore** 2022 - 2026

- GPA: 4.59/5.00
- Focus areas: Programming Language, Parallel Computing
- Courses: CS2106 Operating Systems (A), CS3210 Parallel Computing (A), CS3211 Concurrent Programming (A), CS4212 Compiler Design (A-), CS5223 Distributed Systems

Co-Curricular Activities

President [Computing for Voluntary Welfare Organisations](#) 09/2023 - 08/2024

Oversee the recruitment and application process for CVWO's 2024 Summer Internship program. Manage marketing, outreach, and operations for CVWO.

Backend Tech Lead [NUS Students' NUS College Club](#) 12/2023 - 04/2024

Designed and implemented Role-Based Access Control for the for backend of the NUSC Club website

Skills

- Programming Languages:** Rust, Go, TypeScript, Python
- Databases:** SQL, Redis, Vector DBs
- Tools/Frameworks:** React, SolidJS